How long can a two-cycle quantum conveyor operate?

ResearchPhO — Theoretical examination **30 points**

General instructions

1. The theoretical examination lasts 5 hours. It has three problems; each problem is worth 10 points.
2. You may use a non-programmable scientific calculator. No other aids are allowed.
3. Each problem has its own Answer Sheets. Write each final answer in the box with the same label (for example A.1) on the Answer Sheet of that problem. Use blank writing sheets for derivations you want graded, and mark them with the problem number and the question label.
4. Write on the front side of the sheets only. Use as few words as possible: express yourself with equations, numbers, symbols and diagrams.
5. The problems are independent, and they can be solved in any order. Within a problem, later questions can often be answered without the earlier results.
6. Unless stated otherwise, set $\hbar = 1$. Charge is counted as a number of particles, and transport to the right is positive.

Problems

Problem	Title and parts	Points
1	A two-cycle quantum conveyor: bands, projectors and flavors (Parts A–D)	10.0
2	Correlations and the physical current (Parts A–D)	10.0
3	How long can the conveyor operate? (Parts A–C)	10.0
		30.0

Problem 1: A two-cycle quantum conveyor: bands, projectors and flavors

(10.0 points)

In a *Thouless pump* the parameters of a lattice Hamiltonian are changed slowly and periodically in time. After each period T the Hamiltonian is the same as before, yet particles can be carried along the lattice. For non-interacting fermions that completely fill a band separated by gaps from the other bands, the number of particles carried past any bond in one period is an integer, the Chern number of the band.[1]

This problem studies spinless fermions on a chain with five sites per unit cell (Fig. 1.1). There are N cells, $5N$ sites and N particles, one particle per cell on average. The Hamiltonian of this "multi-band fractional pump" (MFP) is

$$H_{\mathrm{MFP}}(t) = -\sum_j \left[J_j(t) c_{j+1}^\dagger c_j + \mathrm{h.c.}\right] + \sum_j \sum_{r=1}^{3} U_r n_j n_{j+r},$$

$$J_j(t) = J + K\cos\left(\frac{2\pi t}{T} + \frac{4\pi j}{5}\right), \qquad n_j = c_j^\dagger c_j,$$

where c_j removes a fermion from site j and $U_r > 0$ is the repulsion between two particles r sites apart. The ideal motion studied here is to the left; the target transport is $Q_* = -1/2$ particle per period.

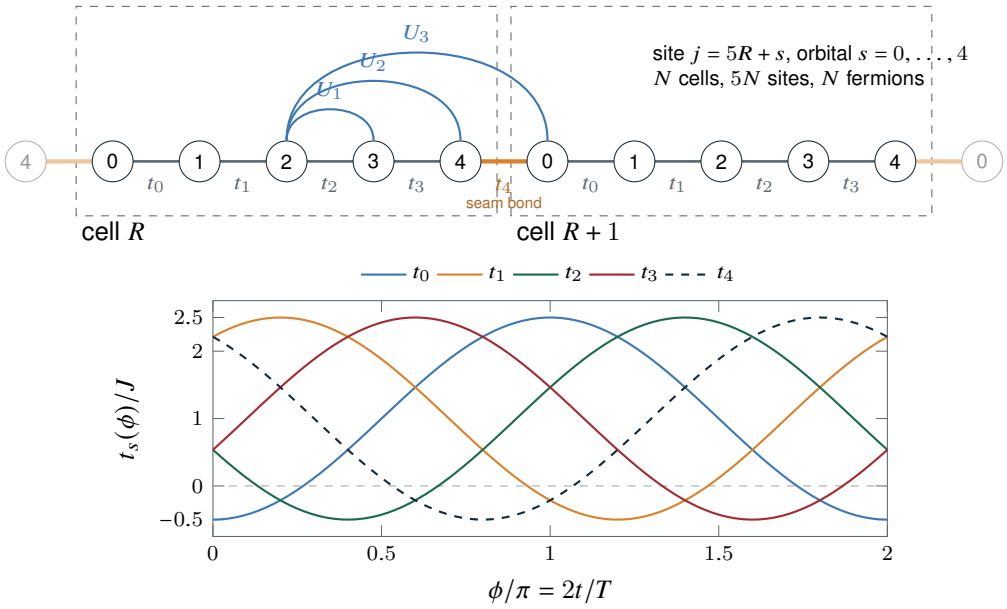


Figure 1.1. The driven chain. Top: site $j = 5R + s$ is orbital s of cell R. The bond from orbital s to $s+1$ has hopping $t_s(\phi)$; the bond t_4 (orange) joins orbital 4 of cell R to orbital 0 of cell $R+1$. Blue arcs: repulsion U_r between particles $r = 1, 2, 3$ sites apart. Bottom: the five hoppings at $K/J = -3/2$ during one period, $\phi = 2\pi t/T$.

Without interactions, translation by one cell lets one write the one-particle problem as a 5×5 matrix $h(k, \phi)$ for each quasi-momentum k; its five eigenvalues are the bands (Fig. 1.2). From all the Bloch states of one band, or of a group of bands, one can build orbitals that are localized in space, one per cell for each band (*Wannier orbitals*). The expectation value of the position in such an orbital is its *centre*. If every orbital of a set is occupied and every centre moves by Δx, the transported charge per cell is $\Delta x/a$, where a is the cell length. Orbitals built from band 1 alone move by $+2$ cells per period (Fig. 1.3a). Mixing bands 1 and 2 gives two localized orbitals per cell, called flavor A and flavor B (Fig. 1.3b). A strong finite-range repulsion can make the particles occupy one flavor only.

[1] This problem is based on M. Jürgensen, J. Steiner, G. Refael and M. C. Rechtsman, arXiv:2504.09338, and S.-M. Huang, arXiv:2608.06777.

The papers behind this problem ask two questions. Can repulsion turn this integer pump into one that carries a fraction of a particle per period, although there is one particle per cell? And how can a moving group of bands be described without choosing a basis inside it (a “projector” description)?

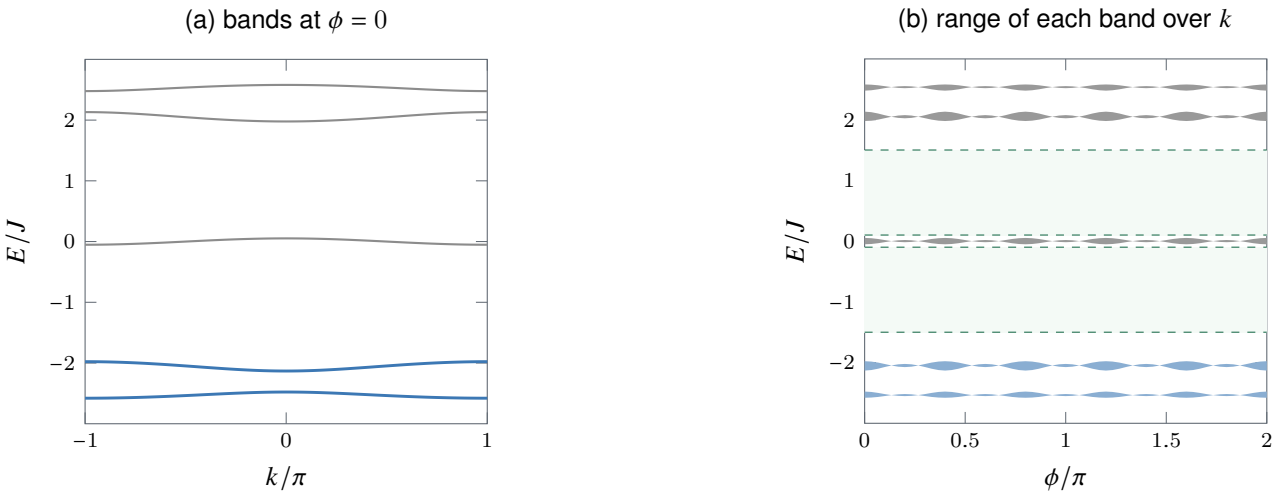


Figure 1.2. Single-particle bands at $K/J = -3/2$. (a) The five bands at $\phi = 0$. (b) The range of each band over all k as ϕ runs over one period. Blue: bands 1 and 2; grey: bands 3–5. No eigenvalue lies in the shaded energy windows $[-3/2, -1/10]$ and $[1/10, 3/2]$.

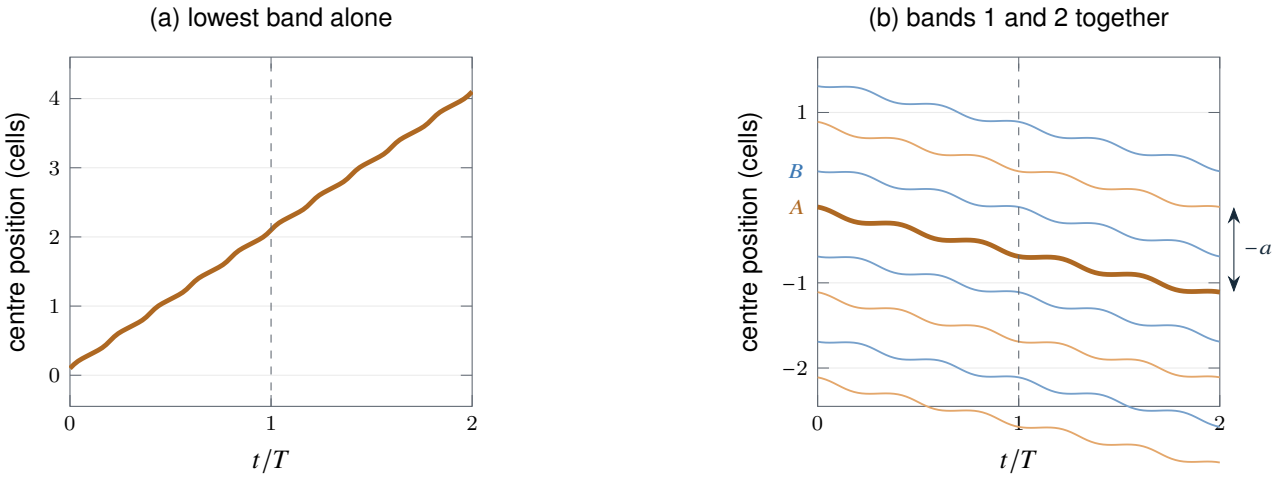


Figure 1.3. Centres of localized orbitals. Computed from $h(k, \phi)$ at $K/J = -3/2$ (single-particle geometry, 24-cell ring). (a) Orbitals built from band 1 alone. (b) Orbitals built from bands 1 and 2 together; orange curves start on flavor-A positions and blue curves on flavor-B positions at $t = 0$. The thick curve is one followed orbital; after $2T$ it is displaced by $-a$.

Useful information

- An orthogonal projector obeys $p^2 = p = p^\dagger$; its rank is $\mathrm{Tr}\, p$.
- On a ring of length L, let P project onto the occupied one-particle states of a group of bands. For a long ring, the eigenvalues of $Pe^{2\pi i x/L}P$ on the range of P are close to $e^{2\pi i x_\alpha/L}$, where x_α are the centres of the maximally localized orbitals of the group. In momentum space this operator is the *Wilson loop*: the ordered product of the overlaps of the band eigenvectors around the Brillouin zone at fixed ϕ.
- For a real unit vector $\boldsymbol{d}$ and real x: $e^{ix\boldsymbol{d}\cdot\boldsymbol{\sigma}} = \mathbb{1}\cos x + i(\boldsymbol{d}\cdot\boldsymbol{\sigma})\sin x$, where $\mathbb{1}$ is the identity.

Part A. Fractional transport (1.5 points)

Suppose that the occupied orbitals follow one flavor exactly. An N-cell ring has cell length a and contains N particles. After $2T$, each occupied orbital has an unwrapped centre displacement $\Delta x_j = -a$. Define the spatially averaged transported particle number by

$$Q(0, 2T) = \frac{1}{Na} \sum_{j=1}^{N} \Delta x_j.$$

The single-period values $Q(0, T)$ and $Q(T, 2T)$ are defined in the same way from the displacements during each period.

A.1 Find $Q(0, 2T)$ and the transport per drive period, averaged over these two periods. Must the two single-period values each equal $-1/2$? Explain. **1.5 pt**

Part B. From the lattice to a projector (4.1 points)

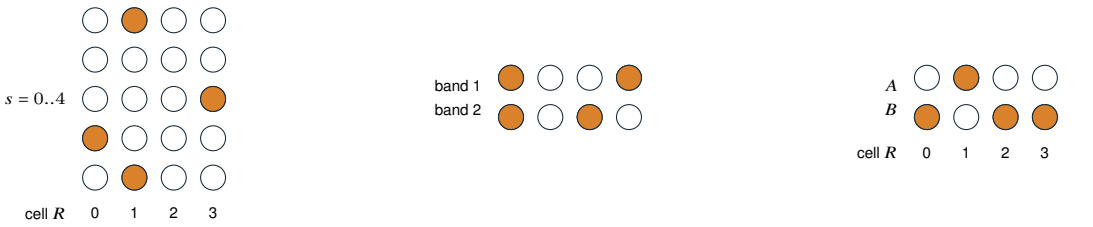
Write the site index as $j = 5R + s$, with cell R and orbital $s = 0, \ldots, 4$. Define

$$t_s(\phi) = J + K\cos(\phi + 4\pi s/5), \qquad c_{R,s} = \frac{1}{\sqrt{N}} \sum_k e^{ikR} c_{k,s},$$

where $k = 2\pi m/N$ is a dimensionless cell momentum. The single-particle part of H_{MFP} is

$$H_0 = -\sum_R \left[\sum_{s=0}^{3} t_s c^\dagger_{R,s+1} c_{R,s} + t_4 c^\dagger_{R+1,0} c_{R,4} + \mathrm{h.c.} \right],$$

and $H_{\mathrm{MFP}} = H_0 + H_{\mathrm{int}}$. Choosing N occupied modes among M distinct fermionic modes gives $\binom{M}{N}$ occupation states. Exactly one occupied flavor per cell is an additional restriction, not the same as average filling one.



(i) any 4 of the $5N$ site orbitals

(ii) any 4 of the $2N$ two-band modes

(iii) exactly one of A, B in every cell

Figure 1.4. Three ways to place $N = 4$ particles in $N = 4$ cells. (i) Any 4 of the $5N$ site orbitals. (ii) Any 4 of the $2N$ two-band modes. (iii) Exactly one of A, B in every cell. Filled circles show one example pattern each.

B.1 **(a)** Derive the 5×5 matrix $h(k, \phi)$ in the ordered basis $(0, 1, 2, 3, 4)$. Check both complex seam entries and Hermiticity for real k. **2.7 pt**
(b) Let $Y = (|u_1\rangle, |u_2\rangle)$ contain two orthonormal lowest-band eigenvectors. Construct $p_{12} = YY^\dagger$. Find its rank and ambient dimension. Why is this object physically motivated in MFP?
(c) Compare $\binom{5N}{N}$, $\binom{2N}{N}$, and 2^N. State the different restrictions behind them. Evaluate all three at $N = 4$. Has Fourier transformation solved the many-body problem?

Hint. Insert the Fourier expansion into each term of H_0 and use $\frac{1}{N}\sum_R e^{i(k'-k)R} = \delta_{kk'}$. Write $H_0 = \sum_k \boldsymbol{c}_k^\dagger h(k,\phi)\boldsymbol{c}_k$ with $\boldsymbol{c}_k = (c_{k,0},\dots,c_{k,4})^T$.

A projector represents a subspace without choosing a basis inside it. Let $p(t)$ be a smooth orthogonal projector, so $p^2 = p = p^\dagger$.

B.2 Show that $p\dot{p}\,p = 0$. Define $g_t = i[\dot{p}, p]$ and prove $[g_t, p] = i\dot{p}$. State what this identity determines and what it does not determine about the actual Hamiltonian. **1.4 pt**

Part C. Following a moving projector **(2.3 points)**

Consider one spin in the Hamiltonian

$$H(t) = -\frac{\Delta}{2}\big[\sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big] = -\frac{\Delta}{2}\,\boldsymbol{n}(t)\cdot\boldsymbol{\sigma}, \qquad \Delta, \nu > 0,$$

where $\boldsymbol{n}(t) = (\sin\nu t, 0, \cos\nu t)$ is the direction of the field. A pure state $|\psi\rangle$ of the spin is pictured by its Bloch vector $\langle\psi|\boldsymbol{\sigma}|\psi\rangle$, a unit vector (Fig. 1.5); the ground state of $H(t)$ has Bloch vector $\boldsymbol{n}(t)$. The instantaneous ground-state projector of $H(t)$ and a rotating frame are

$$p(t) = \tfrac{1}{2}\big[\mathbb{1} + \sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big], \qquad W(t) = e^{-i\nu t\,\sigma_y/2}, \qquad |\Xi(t)\rangle = W^\dagger(t)|\psi(t)\rangle.$$

Use $|\psi(0)\rangle = |\uparrow\rangle$, where $\sigma_z|\uparrow\rangle = |\uparrow\rangle$.

$\boldsymbol{n}(t) = (\sin\nu t, 0, \cos\nu t)$
ground-state projector $p(t) = \frac{1}{2}(\mathbb{1} + \boldsymbol{n}\cdot\boldsymbol{\sigma})$
frame $W(t)$ turns the axes with $\boldsymbol{n}(t)$

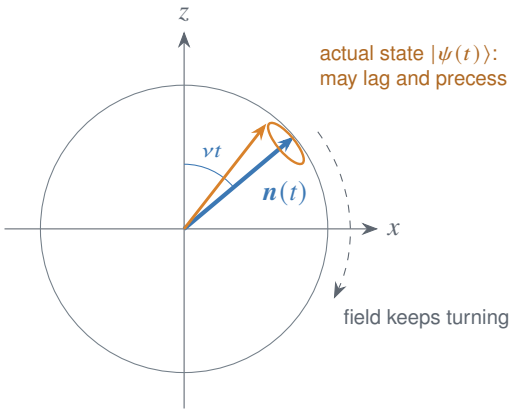


Figure 1.5. A spin in a turning field. Cut through the Bloch sphere in the x–z plane. The field direction $\boldsymbol{n}(t) = (\sin\nu t, 0, \cos\nu t)$ turns at rate ν; the instantaneous ground state points along $\boldsymbol{n}(t)$. The actual state $|\psi(t)\rangle$ may lag and precess.

C.1 Find the time-independent moving-frame Hamiltonian $\mathcal{K} = W^\dagger H W - iW^\dagger\dot{W}$. Find the exact probability $P_{\mathrm{ex}}(t)$ of occupying the instantaneous excited state. Explain how changing Δ can change this probability without changing the projector path. **2.3 pt**

Part D. A plane is not yet a flavor (2.1 points)

Let $Y(k,\phi) = (|u_1\rangle, |u_2\rangle)$ and let $V(k,\phi)$ be a unitary 2×2 matrix with columns v_α. Define

$$u_\alpha = Y v_\alpha, \qquad p_\alpha = Y v_\alpha v_\alpha^\dagger Y^\dagger, \qquad \alpha = A, B.$$

The Wannier orbitals of the two flavors have the form

$$|w_{R,\alpha}\rangle = \frac{1}{\sqrt{N}} \sum_{k,n=1,2} e^{-ikR} V_{n\alpha}(k,\phi) |\psi_{kn}(\phi)\rangle,$$

where $|\psi_{kn}\rangle$ is a full Bloch state; the physical centre uses the actual site positions $x_{R,s} = Ra + r_s$. In a two-dimensional space with labelled directions A and B, compare the two internal maps $\mathcal{T}_0 = \mathbb{1}_2$ and $\mathcal{T}_1 = i\sigma_x$; no real-space displacement is specified for them.

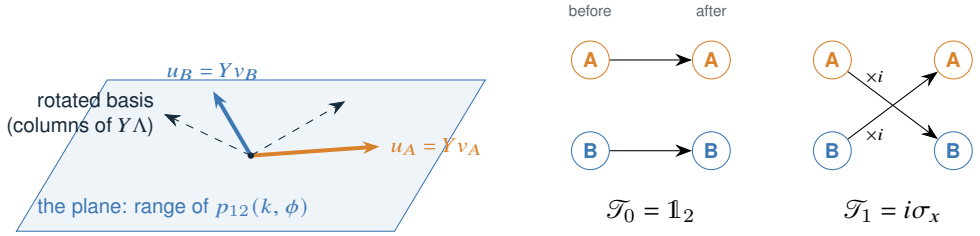


Figure 1.6. A plane and two directions in it. Left: the range of p_{12} at one (k,ϕ), the flavor directions u_A, u_B, and a rotated basis. Right: the two internal maps $\mathcal{T}_0 = \mathbb{1}_2$ and $\mathcal{T}_1 = i\sigma_x$ acting on the labelled directions A and B.

D.1 **(a)** Show that replacing Y by $Y\Lambda$ for any unitary Λ leaves p_{12} unchanged. Show that a fixed physical p_A also stays unchanged if $v_A \to \Lambda^\dagger v_A$. What changes if one selects a different internal direction without this compensation? **2.1 pt**
(b) Explain what must be evaluated to obtain the centers $\langle w_{R,\alpha}|x|w_{R,\alpha}\rangle$ and a continuously labelled A/B path. Distinguish the Wilson loop around k from evolution over physical time.
(c) Use the two maps $\mathcal{T}_0$ and $\mathcal{T}_1$ defined above. Does a closed composite projector or a determinant phase determine internal exchange? Does internal exchange alone prove charge transport?

Problem 2: Correlations and the physical current (10.0 points)

This problem uses the driven chain of Problem 1 but can be solved on its own. Spinless fermions hop on a chain with five sites per cell, one particle per cell on average; the hoppings are modulated with period T and particles $r = 1, 2, 3$ sites apart repel each other:

$$H_{\mathrm{MFP}}(t) = -\sum_j \left[J_j(t) c_{j+1}^\dagger c_j + \mathrm{h.c.}\right] + \sum_j \sum_{r=1}^{3} U_r n_j n_{j+r}.$$

In the ideal "fractional" regime every particle occupies one of two localized orbital types (flavors A or B) and follows it, which transports $-1/2$ particle per period. In a long chain, regions of the other flavor (*domains*) can appear. A practical description therefore starts from a simple *reference state*, transforms the Hamiltonian to frames in which that reference is simple, keeps part of the transformed generator, and bounds the rest. This problem examines that strategy with exactly solvable models and exact identities. Its question is: when a reference state is dressed by correlations and followed in a moving frame, which steps are exact, where does the first approximation enter, and how is the physical current obtained?

Useful information

- If $i\partial_t|\psi\rangle = H|\psi\rangle$ and $|\xi\rangle = U^\dagger|\psi\rangle$ with a differentiable unitary $U(t)$, then $i\partial_t|\xi\rangle = (U^\dagger H U - iU^\dagger \dot{U})|\xi\rangle$.
- If $i\partial_t|\xi\rangle = \mathcal{K}|\xi\rangle$ and $O(t)$ is an operator, then $\frac{\mathrm{d}}{\mathrm{d}t}\langle\xi|O|\xi\rangle = \langle\xi|\big(\partial_t O + i[\mathcal{K}, O]\big)|\xi\rangle$.
- Pauli matrices: $\sigma_a\sigma_b = \delta_{ab}\mathbb{1} + i\epsilon_{abc}\sigma_c$, e.g. $[\sigma_z, \sigma_x] = 2i\sigma_y$; operators on different spins commute. $\mathbb{1}$ is the identity.
- A *Slater determinant* is a state of independent fermions in chosen orbitals; a single (double) excitation replaces one (two) of its occupied orbitals.

Part A. Domains in a spin chain (1.3 points)

Consider the spin chain

$$H_{\mathrm{d}} = -J_z \sum_j \sigma_j^z \sigma_{j+1}^z - h \sum_j \sigma_j^z - g \sum_j \sigma_j^x, \qquad J_z, h > 0.$$

Let A mean $\sigma^z = -1$ and B mean $\sigma^z = +1$. Set $g = 0$. Start with all spins in A. Flip a consecutive segment of ℓ spins to B, with $1 \le \ell < N$. There are two interfaces between the segment and its surroundings (Fig. 2.1).

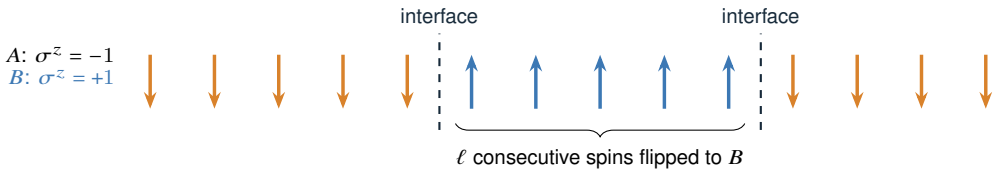


Figure 2.1. A domain in the spin chain. Orange: A ($\sigma^z = -1$). Blue: B ($\sigma^z = +1$). A segment of ℓ consecutive spins is flipped to B; it has two interfaces with its surroundings.

A.1 Find the energy change $\Delta E(\ell)$. Find the smallest integer ℓ for which it is strictly negative, provided that segment fits in the chain. Does an isolated system initially in the all-A state then have to form this lower-energy domain when $g = 0$? **1.3 pt**

Part B. Dressing a reference state (2.6 points)

Retain only a bare two-cell reference $|0\rangle = |o, o\rangle$, with both particles in the reference orbital o, and one double excitation $|d\rangle$ (Fig. 2.2):

$$H_X = \begin{pmatrix} 0 & v \\ v & \Delta \end{pmatrix}, \qquad v, \Delta > 0, \qquad \text{basis } (|0\rangle, |d\rangle).$$

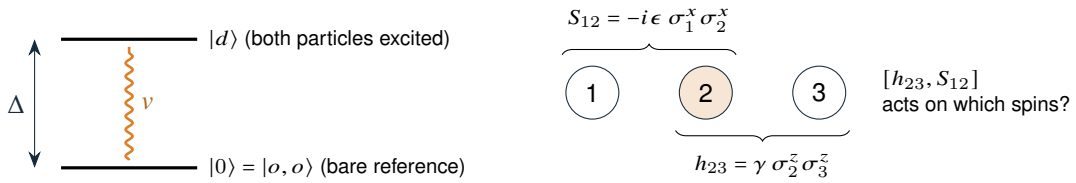


Figure 2.2. A two-level cluster and two overlapping operators. Left: the bare state $|0\rangle$ couples with strength v to $|d\rangle$, which lies Δ higher. Right: S_{12} acts on spins 1 and 2, h_{23} on spins 2 and 3; spin 2 is shared.

B.1 Find the lower eigenvalue and a normalized lower eigenstate. Give their leading forms for $v/\Delta \ll 1$. Starting from the bare state $|0\rangle$, find the exact double-excitation probability $P_d(t)$ and its short-time form. Does this probability define an irreversible decay rate? **1.8 pt**

To see what happens when corrections on neighbouring clusters overlap, consider three spins and

$$h_{23} = \gamma\, \sigma_2^z \sigma_3^z, \qquad S_{12} = -i\epsilon\, \sigma_1^x \sigma_2^x, \qquad \gamma, \epsilon \in \mathbb{R}.$$

B.2 Calculate $[h_{23}, S_{12}]$. Explain why a smaller residual on each isolated pair of cells does not, by itself, certify a small residual after the corrections are assembled along a chain. **0.8 pt**

Part C. Exact frames and kept dynamics (3.1 points)

A particle-number-conserving one-particle frame lifts to the complete many-body Fock space; it does not delete modes or replace interactions by a mean field. A two-body interaction can still produce single and double excitations of a bare Slater reference. Let $W(t)$ be such a differentiable many-body frame and set

$$\mathcal{K}_W = W^\dagger H_{\mathrm{MFP}} W - i W^\dagger \dot{W}.$$

For a differentiable anti-Hermitian many-body $S(t)$, let $\mathcal{U} = We^S$ and $|\zeta\rangle = \mathcal{U}^\dagger |\Psi\rangle$ (Fig. 2.3). Use the identity

$$e^{-S} \partial_t e^S = \int_0^1 e^{-uS} \dot{S} e^{uS}\, \mathrm{d}u.$$

C.1 **(a)** Starting with $i\dot{\Psi} = H_{\mathrm{MFP}}\Psi$, derive the exact generator $\mathcal{K}_{\mathrm{corr}}$ of $|\zeta\rangle$ in terms of $\mathcal{K}_W, S, \dot{S}$. **1.8 pt**
(b) Identify the first-order terms if both S and its time derivative belong to a controlled perturbative expansion. Why may $\dot{S}$ not simply be removed?
(c) If $\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}$, where does an approximation first occur? Write the transformed physical initial state. Explain why dropping $\mathcal{R}$ needs a bound.
(d) Does the one-particle frame turn H_{int} into a one-particle operator? Does writing e^S specify its coefficients or prove that it is local?

Hint. For (b), expand $e^{-S}\mathcal{K}_W e^S = \mathcal{K}_W + [\mathcal{K}_W, S] + \frac{1}{2}[[\mathcal{K}_W, S], S] + \cdots$.

For a fixed normalized direction $|A_0\rangle$ in the chosen frame, let $\Pi_A = |A_0\rangle\langle A_0|$, and write $\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}$ as in C.1. For part (b) use the spin chain of Part A on a ring of N spins, now with constant $g \neq 0$. Set $E_A = \langle A_0|H_{\mathrm{d}}|A_0\rangle$ and let $|j\rangle$ denote the orthonormal configuration with only spin j flipped. You may use $e^{-iHt} = \mathbb{1} - iHt + O(t^2)$. A reference evolution may have the form $|\Psi_{\mathrm{ref}}(t)\rangle = W(t)e^{S(t)}|\Xi(t)\rangle$ with $i\dot{\Xi} = \mathcal{K}_{\mathrm{ret}}(t)\Xi$.

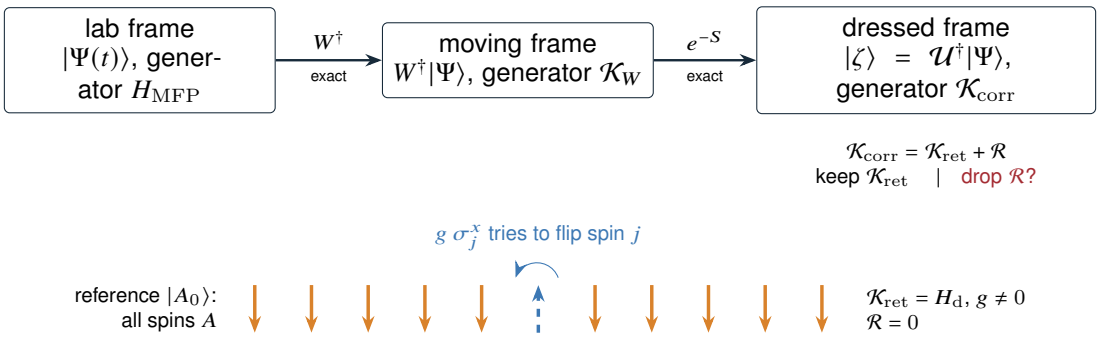


Figure 2.3. Changes of frame. Top: the lab frame, the moving frame W and the dressed frame $\mathcal{U} = We^S$; the generator in the dressed frame is split as $\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}$. Bottom: the all-$A$ reference state of the spin chain; the transverse field g acts on every spin.

C.2 **(a)** Give the necessary and sufficient instantaneous condition for $\mathcal{K}_{\mathrm{ret}}(t)$ to preserve $|A_0\rangle$ up to a phase throughout the interval. **1.3 pt**
(b) Return to the teaching spin chain of Part A, now with constant $g \neq 0$. Take $\mathcal{K}_{\mathrm{ret}} = H_{\mathrm{d}}$ and $\mathcal{R} = 0$. From the all-A state, compute $(H_{\mathrm{d}} - E_A)|A_0\rangle$, its squared norm, and the leading short-time term of $\langle q_j(t)\rangle$, where $q_j = (1 + \sigma_j^z)/2$.
(c) Has the retained model an approximation error in this example? Does zero omitted remainder prove that the reference direction is preserved? Does the short-time spin result give an MFP transport lifetime?

Part D. The physical current

(3.0 points)

Let $|0\rangle$ and $|e\rangle$ be two orthogonal states connected by the current operator

$$I = \lambda \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad |\psi\rangle = \sqrt{1-p}|0\rangle + e^{i\beta}\sqrt{p}|e\rangle, \quad \lambda \in \mathbb{R}, \quad 0 \leq p \leq 1.$$

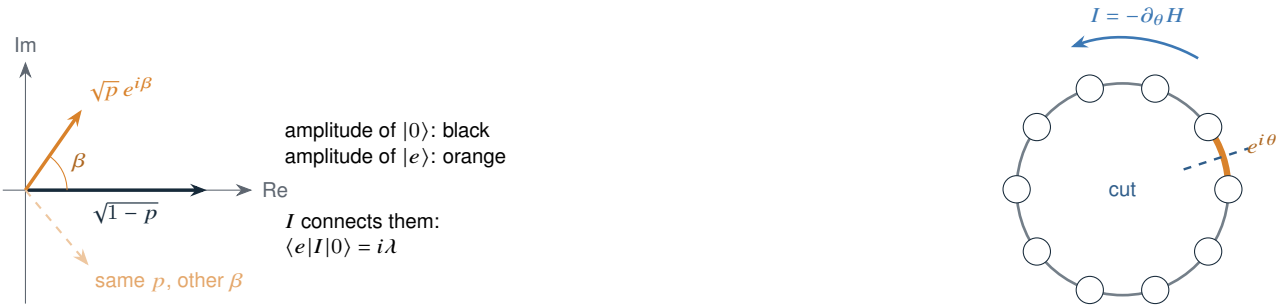


Figure 2.4. Amplitudes and a twisted bond. Left: the amplitudes of $|0\rangle$ and $|e\rangle$ in the complex plane; β is their relative phase. Right: a ring with the phase $e^{i\theta}$ on one bond; $I = -\partial_\theta H$ is the current through that bond.

D.1 Find $\langle I\rangle$ and compare $\beta = 0, \pi/2, -\pi/2$. At small p, is the leading magnitude proportional to p or to $\sqrt{p}$? Can the probability p alone determine the current? **0.9 pt**

Introduce a boundary twist θ: multiply the hopping on one chosen bond, from j to $j+1$, by $e^{i\theta}$ (and its conjugate by $e^{-i\theta}$). Then $I = -\partial_\theta H$ at $\theta = 0$ is the particle current through that bond. The current must be transformed with the same unitary as the Hamiltonian. For a differentiable unitary $\mathcal{U}(\theta, t)$, define

$$\mathcal{A}_\mu = i\mathcal{U}^\dagger \partial_\mu \mathcal{U}, \qquad \mathcal{K} = \mathcal{U}^\dagger H \mathcal{U} - \mathcal{A}_t, \qquad \mu \in \{\theta, t\},$$

and use $\partial_\theta \mathcal{A}_t - \partial_t \mathcal{A}_\theta = i[\mathcal{A}_\theta, \mathcal{A}_t]$. Work on one smooth gauge patch.

D.2 Prove that $\mathcal{U}^\dagger I \mathcal{U} = -\partial_\theta \mathcal{K} - \partial_t \mathcal{A}_\theta - i[\mathcal{K}, \mathcal{A}_\theta]$. For a state satisfying $i\partial_t|\Xi\rangle = \mathcal{K}|\Xi\rangle$, derive the integrated current from t_1 to t_2. Explain why a closed projector path does not justify deleting the endpoint term. **2.1 pt**

Problem 3: How long can the conveyor operate? (10.0 points)

The ideal fractional pump of Problems 1 and 2 transports $Q_* = -1/2$ particle per drive period T (transport to the right is positive). In a real, large, interacting chain the transported charge slowly drifts away from Q_*. This problem builds a method that certifies, window by window, that the charge stays within a tolerance δ. Its question is the open question of the first paper: for how long can one guarantee that the transported charge stays within δ of Q_*, and what would be needed to predict when it fails? It can be solved on its own.

Throughout, ρ' denotes the exact state written in some moving frame, σ' a simpler reference state in the same frame, and I' the physical current operator written in that frame. A region X is a group of cells around the bond where the current is measured (Fig. 3.1); $\bar{X}$ is the rest of the chain.

Useful information

- $\|A\|$ is the operator norm and $\|A\|_1$ the trace norm (sum of singular values); $|\operatorname{Tr}(AB)| \le \|A\|_1 \, \|B\|$. For density matrices, $\|\rho - \sigma\|_1 \le 2$ and $\|\rho - |a\rangle\langle a|\|_1 \le 2\sqrt{1 - \langle a|\rho|a\rangle}$.
- The reduced state on X is $\rho_X = \operatorname{Tr}_{\bar{X}} \rho$ (trace over $\bar{X}$ only). If O_X acts only on X, then $\operatorname{Tr}(\rho O_X) = \operatorname{Tr}_X(\rho_X O_X)$ and $\operatorname{Tr}_{\bar{X}}(O_X \rho) = O_X \rho_X$. For $B_{\bar{X}}$ acting only on $\bar{X}$, $\operatorname{Tr}_{\bar{X}}(B_{\bar{X}} A) = \operatorname{Tr}_{\bar{X}}(A B_{\bar{X}})$.
- For commuting projectors, $1 - \prod_{x \in X}(1 - q_x) \le \sum_{x \in X} q_x$.

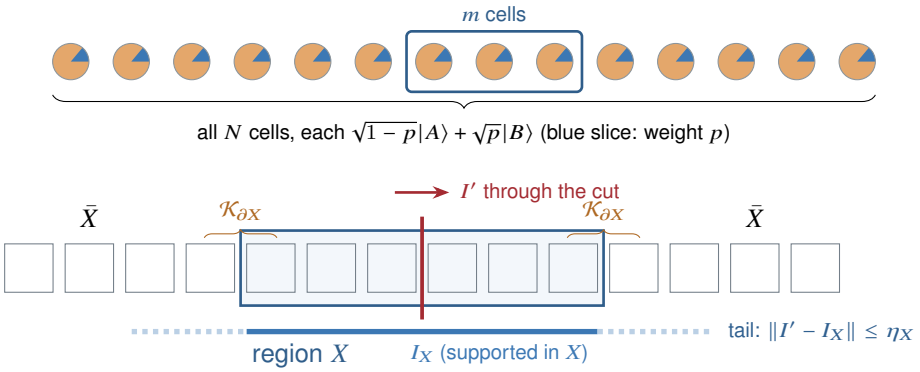


Figure 3.1. A product state and a region around a cut. Top: N cells, each in $\sqrt{1-p}|A\rangle + \sqrt{p}|B\rangle$, and a window of m cells. Bottom: the region X around the cut where the current is measured; $\mathcal{K}_{\partial X}$ couples X to the outside $\bar{X}$; I_X is supported in X.

Part A. Local accuracy in a large system (5.3 points)

Consider the product state

$$|\psi_N\rangle = \bigotimes_{j=1}^{N} \left[\sqrt{1-p}|A\rangle_j + \sqrt{p}|B\rangle_j\right], \qquad 0 < p < 1, \quad \langle A|B\rangle = 0,$$

with p independent of N, and use the fidelity $\mathcal{F}(\rho, |a\rangle) = \langle a|\rho|a\rangle$.

A.1 Find its fidelity with $|A\rangle^{\otimes N}$. Then find the fidelity of an m-cell subsystem with $|A\rangle^{\otimes m}$, with m fixed as N increases. What does this comparison imply about using global fidelity to diagnose local transport? **1.1 pt**

Let ρ' and σ' be normalized density matrices in the same moving frame. Let I_X be supported in X, and suppose $\|I' - I_X\| \le \eta_X$. Reduced density matrices on X are denoted by ρ'_X and σ'_X. Set $\Delta I = \operatorname{Tr}[(\rho' - \sigma')I']$.

A.2 Prove $|\Delta I| \le \|I_X\| \left\|\rho'_X - \sigma'_X\right\|_1 + 2\eta_X$. If the reference is a product state in this frame, define $q_x = 1 - |A_{0,x}\rangle\langle A_{0,x}|$ and $D_X = \sum_{x\in X}\langle q_x\rangle_{\rho'}$. Deduce $|\Delta I| \le 2\,\|I_X\| \min(1, \sqrt{D_X}) + 2\eta_X$. State which form remains valid for a general correlated reference. **2.2 pt**

Now let the two states evolve as

$$i\dot\rho' = [\mathcal{K}_{\rm corr}, \rho'], \qquad i\dot\sigma' = [\mathcal{K}_{\rm ret}, \sigma'], \qquad \mathcal{K}_{\rm corr} = \mathcal{K}_{\rm ret} + \mathcal{R},$$

where $\mathcal{K}_{\rm corr}$ is the exact generator in the chosen frame and $\mathcal{K}_{\rm ret}$ the part kept in the reference description. Both states are normalized; their initial values need not coincide. Write

$$\mathcal{K}_{\rm ret} = \mathcal{K}_X + \mathcal{K}_{\bar X} + \mathcal{K}_{\partial X}, \qquad \sigma'_X = \mathrm{Tr}_{\bar X}\,\sigma',$$

where the first two terms act only inside and only outside X, and the third contains the couplings between both regions. (The "composite-band gap" is the single-particle gap between bands 2 and 3 of the chain, Problem 1.)

A.3 **(a)** Derive the equation for σ'_X. Show why the contribution from $\mathcal{K}_{\bar X}$ vanishes under the partial trace. **2.0 pt**
(b) Is $\sigma'_X(0)$ and $\mathcal{K}_X(t)$ enough to solve that equation? Identify the additional object required by the boundary term. Does the composite-band gap supply it?
(c) Let $\delta\rho = \rho' - \sigma'$. Derive its exact equation and identify the initial-mismatch and omitted-dynamics inputs. Explain why rewriting this equation is not yet a bound on $\|\delta\rho_X(t)\|_1$.

Part B. A conditional operating-time certificate (1.6 points)

Let $\alpha, I_0 > 0$ and $\eta, \delta_0 \ge 0$. Assume that, for every $0 \le s \le t_{\max}$,

$$D_X(s) \le \alpha^2 s^2, \qquad \|I_X(s)\| \le I_0, \qquad \eta_X(s) \le \eta,$$

and that every relevant two-period reference window satisfies $|\overline{Q}_{\rm ref}(t) - Q_*| \le \delta_0$. Here α, I_0 and η have units of inverse time; δ_0 and charge are dimensionless. Define

$$\overline{Q}(t) = \frac{1}{2}\int_{t-2T}^{t} \langle I(s)\rangle\,\mathrm{d}s, \qquad 2T \le t \le t_{\max},$$

and $\overline{Q}_{\rm ref}(t)$ in the same way with the reference state. The factor is $1/2$, not $1/(2T)$: this is charge per drive period averaged over two periods.

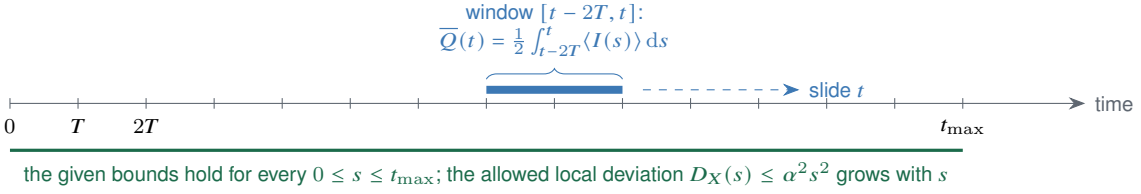


Figure 3.2. A sliding two-period window. Each window $[t-2T, t]$ gives one averaged charge $\overline{Q}(t)$.

B.1 Use A.2 to bound $|\overline{Q}(t) - Q_*|$. For $\alpha I_0 > 0$, find a sufficient upper endpoint t_{cert} for which this bound is at most a tolerance δ. State the positivity and time-domain conditions. Evaluate the result with the illustrative inputs below. Does the next period necessarily fail? **1.6 pt**

Illustrative inputs (time unit $\tau_0 = T$): $I_0\tau_0 = 1$, $\alpha\tau_0 = 10^{-5}$, $\eta\tau_0 = 10^{-5}$, $\delta_0 = 0.001$, $\delta = 0.005$, $t_{\max} = 201\tau_0$.

Part C. Prediction, error and first failure (3.1 points)

For the same frame and the same physical current, A.2 gives $|\Delta I(s)| \le \|I_X(s)\| \left\|\rho'_X(s) - \sigma'_X(s)\right\|_1 + 2\eta_X(s)$. A reference charge must be calculated from its state. Define $\widehat{Q}(t) = \frac{1}{2}\int_{t-2T}^{t} \mathrm{Tr}[\sigma'(s)I'(s)]\,\mathrm{d}s$, and suppose computable functions satisfy

$$\left\|\rho'_X(s) - \sigma'_X(s)\right\|_1 \le e_X(s), \qquad \|I_X(s)\| \le b_X(s), \qquad \eta_X(s) \le \eta(s).$$

For the arithmetic take $Q_* = -0.5$, $\delta = 0.005$, one-window exceedance (no persistence rule), and three invented sample windows in time order:

Sample window	$\widehat{Q}$	E
1	−0.4970	0.0010
2	−0.4940	0.0015
3	−0.4940	0.0005



Figure 3.3. A prediction with an error bar. The prediction $\widehat{Q}$ for one window with error E (blue bar), and the tolerance band $[Q_* - \delta, Q_* + \delta]$ (grey).

C.1 **(a)** Integrate the current inequality to obtain $|\overline{Q}(t) - \widehat{Q}(t)| \le E(t)$. State where initial mismatch must enter this construction. **1.7 pt**
(b) Derive sufficient tests to certify that a given window is inside or outside the tolerance $|\overline{Q} - Q_*| \le \delta$. What is the output when neither test holds?
(c) Apply the tests to the three supplied sample windows. Is the first failed sampled window determined? Is the first failure over continuously varying window starts determined?

For the original pump one would seek a prediction $\widehat{Q}(t)$ and an error $E(t)$ with $|\overline{Q}(t) - \widehat{Q}(t)| \le E(t)$, starting from $H_{\mathrm{MFP}}(t)$ and a specified physical initial state, valid over complete drive cycles and useful as the chain grows. A window is certified to pass if $|\widehat{Q} - Q_*| + E \le 0.005$ and to fail if $|\widehat{Q} - Q_*| - E > 0.005$.

C.2 List the inputs that must be derived from H_{MFP} and a specified physical initial state before either test can be applied to the original pump. Why does certifying individual windows not, by itself, locate the first failure? **1.4 pt**

End of the theoretical examination